COMPSCI 4NL3: Annotation Setup

Winter 2026
Due: February 13th

1 Overview

Now that you have decided on a task and how to collect your data, it is time to begin with data collection (if required) and annotate your data. You will complete the following steps during this phase:

1. **Collect data:** Collect data that fits within the constraints of your task if you do not already have the data that you need. It should be possible for students in your group to determine the label by reading the data.
2. **Develop annotation guidelines:** Write the set of rules that another person could use to label your data, and follow these rules to label a portion of your data. You should set up a simple interface (or spreadsheet) to annotate the data. How you do this is up to you, but you should end up with a CSV or JSON file with your labeled data.
3. **Annotate Data:** You will annotate your data, having each group member annotate data in two phases, initial annotation and reannotation, and finally calculation of agreement.
4. **Report:** Writing a report on your progress and the process of the annotation design, as well as your findings from the experience.

2 Data Collection

You should now begin collecting your data if you do not already have the data you need. **Do not save this step for the day before the due date.** It may take time to ensure you can get the data that you need. This will look different for every project, but you should aim to store your data in an easy-to-access way. Recommended ways to store the data may be in csv, tsv, or json format, where each row or entry is a data point. While collecting your dataset, pay attention to any terms-of-service from the platforms/sources from which you collect data. If you are using an existing dataset, check the license (if one exists).

Finally, make sure to document the dataset thoroughly. This will be the first section of your written report. This should include details about the source of your data and specifically how it was collected in a way that is detailed enough that someone else could read and follow the same procedure. This file should also clearly explain the format of the dataset (what each file represents, what each row or JSON entry corresponds to). Describe what each instance in the dataset represents (e.g., one movie review, one description of a conflict situation), how many total instances there are, the day(s) when the data was collected, how data was sampled from a larger set, and any potentially missing data.

3 Annotation Guidelines and Interface

Your annotations will be completed by members of your group (though you are free to have any other individuals who are willing to help annotate more data). You need to create both the guidelines and the interface that annotators will use to label the data. Again, if a CSV works for you, that is OK. I personally will sometimes make a simple command line application where I have to enter a letter that represents the label and then it prints the next instance, **but make sure to save every time you give input, so that you don't lose your work if the terminal crashes or something else goes wrong**.

3.1 Guidelines

Your annotation guidelines should be a written document which explains the following to a potential annotator:

1. A concise overview of the annotator's job
2. The set of labels that can be applied with descriptions of their meaning (for each item in a numeric scale, or for each class)
3. Unambiguous rules or criteria that should be followed in order to label the data (what is the thinking process you should go through to label the data)
4. At least 1 example of each label along with an explanation for why that is the correct label (and if helpful, why the other labels would not apply to that instance)

Note that this should be written in a way that it can be understood by the annotators regardless of their background knowledge of your dataset or project. **You may think your labels are obvious but you need to be as clear as possible because others may interpret things differently.** To give you an idea, some examples of annotation guidelines can be found in section A.7 of this paper, section A of this paper, or sections A.2 and A.3 of this paper (this last one is videos but you might still find it helpful to have more example instructions to look at).

If your dataset already has labels, you may choose to deviate from this if you wish, as this stage of the project can be treated as more experimental, and you may revert to the provided labels for the next stage. This is up to your team. If you choose to deviate, you should explain why. Your corpus may also have a corresponding paper that describes the annotation procedure. You should not look at this when designing your own annotation scheme but rather try to come up with your own.

3.2 Interface

You should design an interface that can be used for your annotation task and write complete instructions for how someone could use this interface to label your data. It is recommended that you use a command line interface or spreadsheet to label the data.

4 Annotation

You and each team member should follow your instructions to annotate your data. You may run into strange cases with the annotations where you aren't sure what label to apply. If this is rare, you do not have to change your annotation scheme at this point. You may run into interesting edge cases where labels don't apply exactly the way they were intended. Save these for your report.

You should spend at least 1 hour annotating. If your group decided to annotate an entirely new dataset, you may have to spend more time annotating, as you should aim to have at least 1k labeled instances.

After you are done annotating, a portion of your data should be annotated a second time by a different person. What does this look like in practice? If you had each group member annotate data for an hour, each person annotating 300 different data points with 3 group members, you then have 900 different points annotated. Now take 15% of this amount, or 135 instances and divide this by 3, having each team member annotate another 45 instances **that another person has already annotated**. The result of this is that you will have 900 different data points and 135 annotated by two different group members. We will use these to calculate annotator agreement. If you annotated for a longer time, calculate the estimate of the overlapping set based on roughly 1 hour of annotation time.

Include in your report the total number of instances you annotated and how long it takes to annotate an instance (as calculated as an average over annotators and instances annotated).

4.1 Calculating Agreement

Collect the annotations from your teammates. Compile the data into a single file containing all data points and labels. Next, you will need to compute agreement. Agreement metrics help assess the consistency of annotations. Some commonly used metrics include:

1. Cohen's Kappa: For pairwise agreement between two annotators
2. Fleiss' Kappa: For agreement among multiple annotators
3. Krippendorff's Alpha: Useful when not all annotators annotated all data

These metrics will help you determine if your annotations are reliable or if there are significant disagreements among annotators. Read more about each to see which one is most suitable for your dataset and compute that metric. You may use existing libraries to do this for you. The one that is the most appropriate for most of you will be Krippendorff's Alpha if you split the secondary annotation between 3+ people. If you had only two people annotate duplicate instances, you should likely use Cohen's Kappa.

5 Deliverables

You should submit the following A2L:

1. (10%) Your annotated dataset
2. (90%) A PDF of your report
3. The interface if you use a command line program

Your report should contain the following (with score %s adding up to the 90% for the report as noted above):

1. Your group number and list of members
2. (40%) Annotation instructions, covering all points asked of you in §3.1
3. (10%) A description of the dataset and a disclaimer about any sensitive content that may be contained in the data to be annotated if relevant. The description should follow the guidelines of §2 paragraph 2.
4. (10%) A description of at least 3 interesting data points you found and what makes them interesting, along with the estimates requested in §4. What metric you used for annotator agreement and it's value.
5. (30%) Answers to the following questions:
   (a) What did you learn about the task from doing the annotation?
   (b) What challenges do you expect models to face when learning from your data?
   (c) What surprising things did you observe in the data? (you can refer to the points mentioned from §4 if you wish, but you may have more to say here)
   (d) Which features do you expect to be useful?
   (e) Describe the mental model you came up with to do the task at a high level. What process did you use?
   (f) Was there anything unclear about the instructions? How would you recommend modifying them in the future to make them more helpful?

There is no page limit for the report, but you will likely have 3-5 pages, possibly more depending on how many examples you want to include.